

'Cause We Are Living in a Counterfactual World

Lecture 1: Counterfactual Framework and Selection on Observables

PhD Economics, Management and Methods for the Sustainable Transition • University of Ferrara

Francesco Rentocchini

European Commission, JRC-Seville / DEMM, University of Milan • 5-7 May 2026

Motivation: Why Causality?

The Potential Outcomes Framework

Identification: from Description to Causality

Estimators for Selection on Observables

Lab: Titanic and LaLonde

Motivation: Why Causality?

The Potential Outcomes Framework

Identification: from Description to Causality

Estimators for Selection on Observables

Lab: Titanic and LaLonde

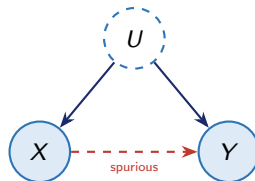
Correlation is not causation — and it matters

- Countries with more storks have higher birth rates
- Ice-cream sales predict drowning deaths
- Hospitals kill more people than battlefields

All three statements are **true correlations** and **useless**.

The policy question

Does *this programme* improve outcomes for the people who receive it? What would have happened to them *without it*?



A common cause U creates a correlation between X and Y even when X has *no* causal effect on Y .

Hume's challenge: we infer causation, we never observe it

Hume, *Treatise* (1739) & *Enquiry* (1748):

- We observe **sequences**: flame touches paper → paper burns
- We observe **regularities**: this always happens
- We *never* observe the necessary connection itself
- “Causal power” is **constant conjunction + habit of expectation** — a feature of our *minds*, not the world

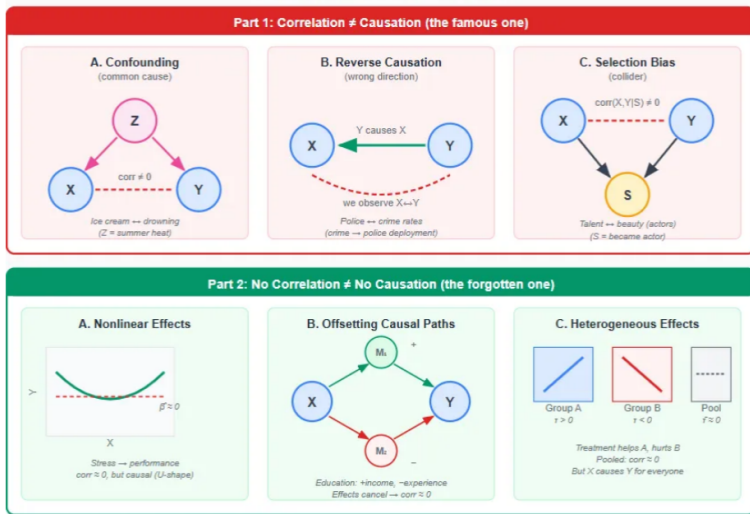
Hume's inescapable lesson

Every causal claim adds assumptions **beyond what the data contain**. The task: make those assumptions explicit.

Three centuries of causal thinking: a roadmap

Who	When	Key move
Hume	1739	Stated the problem: causation is inferred, never directly observed
Mill	1843	Method of difference: compare units differing in one cause only
Galton & Pearson	1880s–1900s	Formalised correlation — powerful tool, but conflated with causation
Fisher	1925	Randomisation: <i>design</i> beats analysis — balance confounders by chance
Neyman / Rubin	1923 / 1974	Potential outcomes $Y_i(1) - Y_i(0)$: put the counterfactual at the centre
Haavelmo / Cowles	1944–55	Structural models + IV: theory-driven identification from observational data
Pearl	1995	DAGs: graphical language for assumptions and identification conditions
Angrist, Card, Krueger	1990s–	Credibility revolution : quasi-experiments, design-based identification

The two-way failure of correlation



Source: Chavez (2026)

The ventilator problem

- ICU observation: ventilated patients die *more* often
- Naive conclusion: **ventilators kill?** — obviously wrong
- Why? Ventilated patients were **already sicker**
- They were selected into treatment *because* of severity

Patient	Ventilated	Survived
Alice	Yes	No
Bob	Yes	No
Carol	No	Yes
Dan	No	Yes
Naive Δ survival		—ve

The fundamental question

What would have happened to ventilated patients had they *not* been ventilated?

That is the **counterfactual** — unobservable by definition.

Causal inference is the science of disciplined guessing.

Alice and Carol are not comparable:
selection into treatment is not random.

Lecture 1 (Today): Selection on observables

We assume all confounders are *measured*. Estimators: matching, IPW, doubly robust.

Lecture 2: Quasi-experiments — DiD

We exploit *variation in timing* of a policy to construct a counterfactual trend.

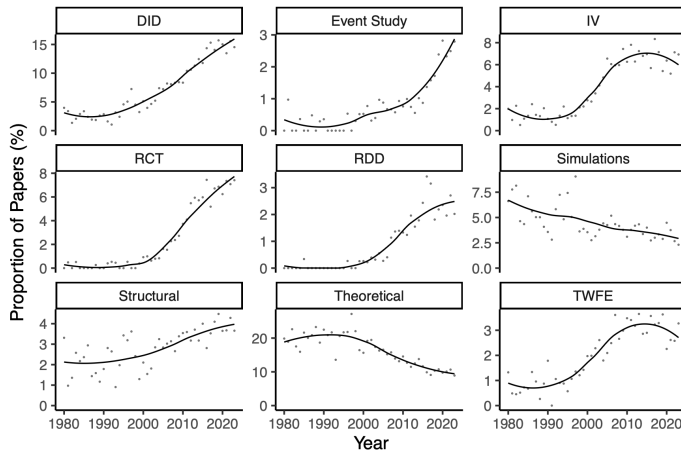
Lecture 3: Quasi-experiments — RDD and Shift-Share IV

We exploit *discontinuities* or *external shocks* to extract exogenous variation.

All three methods answer the same question with different **identifying assumptions**.

The assumption is always a claim about the data-generating process — not about the data itself.

Figure 4: Proliferation of Empirical Methods Over Time in NBER and CEPR Working Papers



Garg and Fetzer (2025)

Motivation: Why Causality?

The Potential Outcomes Framework

Identification: from Description to Causality

Estimators for Selection on Observables

Lab: Titanic and LaLonde

Rubin's potential outcomes notation

Let $i = 1, \dots, N$ index units (firms, individuals, countries).

$D_i \in \{0, 1\}$ Treatment indicator (1 = treated, 0 = control)

$Y_i(1)$ Outcome unit i *would* realise **if treated**

$Y_i(0)$ Outcome unit i *would* realise **if not treated**

$\delta_i = Y_i(1) - Y_i(0)$ **Individual treatment effect**

Switching equation — what we actually observe (that is Y_i):

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

Key insight

$Y_i(1)$ and $Y_i(0)$ both exist in principle for every unit. We observe at most *one* of them. The other is the **counterfactual**.

Potential outcomes: a complete (fictional) table

Unit	D_i	$Y_i(1)$	$Y_i(0)$	τ_i
Alice	1	1	0	1
Bob	1	0	0	0
Carol	0	1	1	0
Dan	0	0	0	0

In a fictional world we *observe both* potential outcomes for every unit.

Exercise

Using the complete table, compute:

- ATE =?
- ATT =? (treated: Alice, Bob)
- ATU =? (untreated: Carol, Dan)

Are they equal? What does that tell you?

Unit	D_i	$Y_i(1)$	$Y_i(0)$	τ_i
Alice	1	1	0	1
Bob	1	0	0	0
Carol	0	1	1	0
Dan	0	0	0	0
ATE				$\frac{1+0+0+0}{4} = \mathbf{0.25}$
ATT (Alice, Bob)				$\frac{1+0}{2} = \mathbf{0.50}$
ATU (Carol, Dan)				$\frac{0+0}{2} = \mathbf{0.00}$

Are they equal?

No: $ATT > ATE > ATU$.

What does that tell you?

Units who *chose* treatment (Alice, Bob) gain more from it than those who did not. This is **positive selection into treatment**.

In the real world we cannot compute any of these directly — we observe only *one* potential outcome per unit. That is why identification strategy matters.

The fundamental problem of causal inference

Unit	D_i	$Y_i(1)$	$Y_i(0)$
Alice	1	1	?
Bob	1	0	?
Carol	0	?	1
Dan	0	?	0

? = never observed. This is a **missing data problem**, always.

Naive solution

Simple difference in outcomes: compares treated to untreated directly

Average treatment effects:

$$ATE = E[Y_i(1) - Y_i(0)]$$

$$ATT = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$

$$ATU = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

Which parameter?

ATT = effect on those who *chose* to be treated. Most policy-relevant when treatment is voluntary.

ATE = effect if we assigned treatment *randomly* to the population.

Selection bias: what goes wrong with naive comparisons

Step 1. The naive estimator compares treated to untreated:

$$\underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{Simple Difference in Outcomes (SDO)}}$$

What does SDO actually measure?

We observe $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$, so:

$$E[Y_i | D_i = 1] = E[Y_i(1) | D_i = 1], \quad E[Y_i | D_i = 0] = E[Y_i(0) | D_i = 0]$$

SDO mixes the treatment effect *and* pre-existing differences.

Selection bias: what goes wrong with naive comparisons

Step 1. The naive estimator compares treated to untreated:

$$\underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{Simple Difference in Outcomes (SDO)}}$$

Step 2. Add and subtract $E[Y_i(0) | D_i = 1]$:

$$\begin{aligned} \text{SDO} &= E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 0] \\ &= E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 1] + E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0] \end{aligned}$$

(Add and subtract the *unobserved* baseline of the treated group.)

Selection bias: what goes wrong with naive comparisons

Step 1. The naive estimator compares treated to untreated:

$$\underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{Simple Difference in Outcomes (SDO)}}$$

Step 3. Rearrange — two terms emerge:

$$\text{SDO} = \underbrace{E[Y_i(1) - Y_i(0) | D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]}_{\text{Selection bias}}$$

Selection bias = difference in *untreated* potential outcomes between the two groups.

Selection bias: what goes wrong with naive comparisons

Step 1. The naive estimator compares treated to untreated:

$$\underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{Simple Difference in Outcomes (SDO)}}$$

Step 4. With heterogeneous effects, SDO also picks up a third term:

$$\begin{aligned} \text{SDO} = & \underbrace{E[Y_i(1) - Y_i(0) | D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]}_{\text{Selection bias}} \\ & + (1 - \pi) \underbrace{(\text{ATT} - \text{ATU})}_{\text{Heterogeneity bias}} \end{aligned}$$

- where π is the share of treated units
- $\text{SDO} = \text{ATT}$ *only if* selection bias = 0 (treatment independent of $Y_i(0)$)
- Intuition: sicker patients $\rightarrow$ hospital; harder workers $\rightarrow$ training; larger firms $\rightarrow$ M&A

All the above rests on the **Stable Unit Treatment Value Assumption** (Rubin 1980):

1. **No interference:** $Y_i(d)$ does not depend on what treatment D_j unit $j \neq i$ receives
 - Violated when: spillovers, general equilibrium effects, peer effects
 - Example: vaccinating my neighbour affects my infection risk
2. **No hidden treatment variation:** there is only one version of each treatment level
 - Violated when: treatment is heterogeneous in practice (e.g., “training” differs across providers)

Keep in mind

SUTVA is an assumption about the world, not the data. If it fails, $Y_i(d)$ is not well-defined and the whole framework breaks down. Always ask: could my treated units affect my controls?

Motivation: Why Causality?

The Potential Outcomes Framework

Identification: from Description to Causality

Estimators for Selection on Observables

Lab: Titanic and LaLonde

If treatment is assigned **randomly**:

$$D_i \perp (Y_i(0), Y_i(1))$$

Then:

$$E[Y_i(0) \mid D_i = 1] = E[Y_i(0) \mid D_i = 0]$$

Selection bias = 0, and SDO = ATE = ATT.

RCT: the gold standard

Random assignment makes the two groups identical in expectation on all characteristics — observed *and* unobserved.

The problem

We often *cannot* randomise: firms self-select into acquiring; workers choose training; countries adopt policies for a reason. We need an *assumption* to substitute for randomisation.

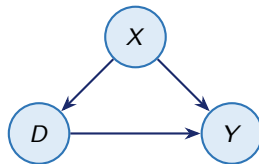
The Conditional Independence Assumption (CIA)

Idea: conditional on observed characteristics X_i , treatment assignment is *as good as random*.

$$(Y_i(0), Y_i(1)) \perp D_i \mid X_i \quad (\text{Unconfoundedness / CIA})$$

What X_i must contain:

- All variables that jointly determine treatment *and* potential outcomes
- I.e., all **confounders** must be measured
- Missing a confounder $\Rightarrow$ CIA fails



Conditioning on X blocks the backdoor $D \leftarrow X \rightarrow Y$

Common support (overlap):

$$0 < P(D_i = 1 \mid X_i) < 1 \quad \forall X_i$$

Every value of X must appear in both treated and control groups. Otherwise, we extrapolate.

A first attempt: just add controls to OLS

Idea: if the bias comes from omitted variables, why not include them?

$$Y_i = \alpha + \beta D_i + \gamma' X_i + \varepsilon_i$$

$\hat{\beta}$ identifies ATT *only if* $E[\varepsilon_i | D_i, X_i] = 0$ — i.e., no residual confounding.

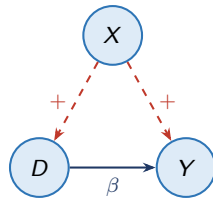
Read the controls as a diagnostic:

Coefficient falls after adding controls

$\hat{\beta}^{\text{naive}} > \hat{\beta}^{\text{OLS}+X}$ — positive selection

Treated units had *better* baseline outcomes: e.g. more able workers self-select into training.

Controls absorb some of the spurious advantage. **Good sign** — you are correcting in the right direction.



X biases $\hat{\beta}^{\text{naive}}$ upward

A first attempt: just add controls to OLS

Idea: if the bias comes from omitted variables, why not include them?

$$Y_i = \alpha + \beta D_i + \gamma' X_i + \varepsilon_i$$

$\hat{\beta}$ identifies ATT *only if* $E[\varepsilon_i \mid D_i, X_i] = 0$ — i.e., no residual confounding.

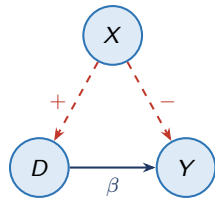
Read the controls as a diagnostic:

Coefficient rises after adding controls

$\hat{\beta}^{\text{naive}} < \hat{\beta}^{\text{OLS}+X}$ — negative selection

Treated units had *worse* baseline outcomes: e.g. sicker patients received the drug.

The naive estimate was attenuated downward. Controls reveal a larger true effect.



X biases $\hat{\beta}^{\text{naive}}$ downward

A first attempt: just add controls to OLS

Idea: if the bias comes from omitted variables, why not include them?

$$Y_i = \alpha + \beta D_i + \gamma' X_i + \varepsilon_i$$

$\hat{\beta}$ identifies ATT *only if* $E[\varepsilon_i | D_i, X_i] = 0$ — i.e., no residual confounding.

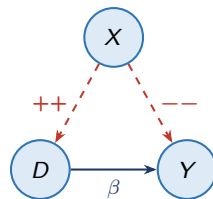
Read the controls as a diagnostic:

Sign flip — the most alarming case

$\hat{\beta}^{\text{naive}} > 0$ but $\hat{\beta}^{\text{OLS}+X} < 0$ (or vice versa). Omitted variable so strongly correlated with D that it *reversed* the raw association.

Classic examples: hospitals $\uparrow$ mortality (naive) $\rightarrow$ negative after controlling for baseline health; elite schools $\uparrow$ wages (naive) $\rightarrow$ near-zero after controlling for ability.

A sign flip is a red flag: OLS with controls may still be insufficient — CIA may not hold.



backdoor overwhelms $\beta \Rightarrow$ sign flip

A first attempt: just add controls to OLS

Idea: if the bias comes from omitted variables, why not include them?

$$Y_i = \alpha + \beta D_i + \gamma' X_i + \varepsilon_i$$

$\hat{\beta}$ identifies ATT *only if* $E[\varepsilon_i \mid D_i, X_i] = 0$ — i.e., no residual confounding.

Read the controls as a diagnostic:

Standard error rises after adding controls

Happens when X is *highly collinear* with D — controls absorb most variation in treatment.

Extreme case: X is a near-perfect predictor of $D \Rightarrow$ **multicollinearity**, very imprecise $\hat{\beta}$.

Also: small within-cell sample sizes inflate SEs even without collinearity.

A first attempt: just add controls to OLS

Idea: if the bias comes from omitted variables, why not include them?

$$Y_i = \alpha + \beta D_i + \gamma' X_i + \varepsilon_i$$

$\hat{\beta}$ identifies ATT *only if* $E[\varepsilon_i \mid D_i, X_i] = 0$ — i.e., no residual confounding.

Read the controls as a diagnostic:

Standard error falls after adding controls

X explains residual variation in Y without being collinear with D .

Controls reduce $\hat{\sigma}_\varepsilon^2 \Rightarrow$ smaller SEs — more precise estimate. This is the *best case*: controls both correct bias *and* improve precision.

Motivation: Why Causality?

The Potential Outcomes Framework

Identification: from Description to Causality

Estimators for Selection on Observables

Lab: Titanic and LaLonde

Idea: partition the covariate space X into K strata. Within each stratum, CIA holds by construction (units are identical on X).

$$\hat{\tau}^{\text{ATE}} = \sum_{k=1}^K \frac{N_k}{N} \cdot \underbrace{(\bar{Y}_k^{D=1} - \bar{Y}_k^{D=0})}_{\text{stratum-specific effect}}$$

Strengths

- Transparent and interpretable
- No functional form imposed
- Naturally handles ATE vs. ATT (different weights)

Weakness

Curse of dimensionality: with p binary covariates, you need 2^p cells. Many will be empty (no overlap).

Solution: collapse X to a scalar — the *propensity score* — or use coarsened matching.

Exact matching: for each treated unit i , find a control unit j with identical $X_j = X_i$.

$$\hat{\delta}_i = Y_i - Y_{j(i)}, \quad \widehat{ATT} = \frac{1}{N_1} \sum_{i:D_i=1} \hat{\delta}_i$$

- Fully non-parametric: no model for $E[Y|X]$
- Estimating is the same as a within-cell comparison
- **Breaks down in high dimensions:** with 10 binary covariates, $2^{10} = 1,024$ cells; few or none will have both treated and control units

The trade-off

Exact matching $\rightarrow$ zero bias, but high variance (few matches, many discarded units).

Inexact / caliper matching $\rightarrow$ lower variance, but introduces bias.

Iacus et al. (2012). **Key idea:** coarsen each covariate into bins, *then* exact-match on the coarsened values.

Algorithm:

1. For each X_k , define bins (e.g., age: 18–25, 26–35, 36–50, 51+)
2. Create a stratum identifier = combination of all coarsened bins
3. Keep only strata that contain *at least one* treated and one control unit
4. Within matched strata, estimate the treatment effect; weight observations by stratum size

Advantages

- Guaranteed covariate balance within strata
- No propensity score model needed
- User controls coarsening $\Rightarrow$ transparent
- Reduces model dependence

Caveats

- Coarsening choices are arbitrary
- Can discard many observations (poor overlap $\Rightarrow$ small matched sample)
- Still relies on CIA: all confounders in X

Propensity score (Rosenbaum & Rubin 1983):

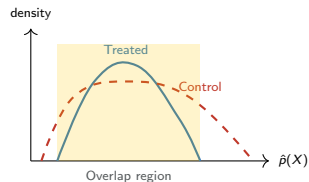
$$p(X_i) = P(D_i = 1 \mid X_i)$$

Balancing property: conditional on $p(X)$, X is independent of D . Matching on a *scalar* $p(X)$ is sufficient to balance all of X .

Steps:

1. Estimate $\hat{p}(X)$ via logit or probit
2. Check **common support**: plot $\hat{p}$ by treatment status; trim tails
3. For each treated unit, find the nearest-neighbour control on $\hat{p}$ (1:1 or 1:k)
4. Estimate ATT as average matched difference

Common support check:



Treated and control should overlap. Trim $\hat{p} < 0.05$ or $\hat{p} > 0.95$.

Idea: re-weight observations so that controls “look like” the treated group in terms of X .

IPW estimator for ATT:

$$\hat{\tau}_{IPW}^{ATT} = \frac{1}{N_1} \sum_{i=1}^N \left[\frac{D_i \cdot Y_i}{1} - \frac{(1 - D_i) \cdot \hat{p}(X_i)}{1 - \hat{p}(X_i)} \cdot Y_i \right]$$

This a weighted OLS, with weights $w_i = D_i + (1 - D_i) \frac{\hat{p}(X_i)}{1 - \hat{p}(X_i)}$.

Intuition

Control units with $\hat{p}$ close to 1 (“almost treated”) are up-weighted. Control units with $\hat{p}$ close to 0 (“nothing like treated”) are down-weighted.

Warning: overlap matters

If $\hat{p}(X_i) \approx 1$ for some controls, weights explode. Always **trim** on propensity score before IPW ($0.1 < \hat{p} < 0.9$ as a starting rule).

Idea: combine an *outcome model* and a *propensity score model*. Consistent if *either* is correctly specified.

$$\hat{\tau}^{\text{DR}} = \frac{1}{N} \sum_i \left[\underbrace{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)}_{\text{outcome model}} + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{p}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{p}(X_i)} \right]$$

where $\hat{\mu}_d(X) = E[Y|D = d, X]$ is the fitted outcome under treatment d .

Idea: combine an *outcome model* and a *propensity score model*. Consistent if *either* is correctly specified.

$$\hat{\tau}^{\text{DR}} = \frac{1}{N} \sum_i \left[\underbrace{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)}_{\text{outcome model}} + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{p}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{p}(X_i)} \right]$$

where $\hat{\mu}_d(X) = E[Y|D = d, X]$ is the fitted outcome under treatment d .

Insurance policy interpretation

If the outcome model is right $\Rightarrow$ the IPW correction term averages to zero $\Rightarrow$ DR is consistent.

If the PS model is right $\Rightarrow$ the outcome model residuals are well-weighted $\Rightarrow$ DR is still consistent.

You don't need both to be right — but you should try hard on both.

Motivation: Why Causality?

The Potential Outcomes Framework

Identification: from Description to Causality

Estimators for Selection on Observables

Lab: Titanic and LaLonde

Dataset 1: the Titanic

Background

RMS Titanic sank on 15 April 1912. Of the 2,224 people on board, 1,502 died. Evacuation followed a “women and children first” protocol; lifeboat access differed sharply by deck.

Data (Dawson, 1995)

Passenger manifest: ticket class (1st/2nd/3rd), sex, age group, survival indicator.

Causal question

Did being a *first-class* passenger *cause* higher survival — or does the raw gap reflect compositional differences in sex and age across classes?

Key results (raw data)

Group	Survival rate
1st class	63%
2nd class	47%
3rd class	24%
Male	19%
Female	74%

The identification problem

1st-class passengers were disproportionately *female* and *adult* — both independently associated with survival. Naive gap conflates class effect with composition.

Dataset 2: LaLonde (1986) and the NSW programme

Background

NSW demonstration (mid-1970s): *randomised* job-training for disadvantaged workers — long-term unemployed, ex-offenders, school drop-outs. Random assignment into training ($D = 1$) or control ($D = 0$).

Data (LaLonde, 1986)

- **Experimental:** NSW treated + NSW controls
- **Observational:** NSW treated + CPS/PSID controls
- Covariates: age, education, race, marital status, earnings 1974–75
- Outcome: real earnings 1978

Experimental benchmark

Training raised earnings by $\approx \$1,794$ (statistically significant).

LaLonde's bombshell (LaLonde, 1986)

Swapping to CPS/PSID controls and re-running OLS and Heckman selection: estimates ranged from $-\$15,000$ to $+\$3,500$.

Observational methods fail when selection is driven by unobservables.

The rehabilitation (Dehejia and Wahba, 2002)

PS matching on the same observational data recovered estimates close to $\$1,794$ — launching the matching literature and the debate on when CIA is credible.

Exercise 1 — Titanic (20 min)

Question: did first-class passengers have higher survival probability?

Problem: first-class passengers differ from others in age, sex — confounders that also affect survival.

Method: subclassification on (sex $\times$ age group), compute ATE, ATT, ATU with explicit weights.

Dataset: `titanic.dta` (Mixtape)

Exercise 2 — LaLonde (70 min)

Question: did the NSW job-training programme raise earnings?

Benchmark: experimental ATT $\approx$ \$1,794 (Lalonde 1986)

Methods: naive OLS $\rightarrow$ IPW $\rightarrow$ PS matching $\rightarrow$ CEM $\rightarrow$ doubly robust

Datasets: `nsw_mixtape.dta` (experimental) + `cps_mixtape.dta` (observational controls)

Script: `Lab/L1_matching.do`

- Chavez, C. (2026). On causality: A history of how economics learned to think about cause and effect. Substack. <https://carloschavezp29.substack.com/p/on-causality>.
- Dawson, R. J. M. (1995). The 'unusual episode' data revisited. *Journal of Statistics Education*, 3(3).
- Dehejia, R. H. and Wahba, S. (2002). Propensity score-matching methods for nonexperimental causal studies. *Review of Economics and Statistics*, 84(1):151–161.
- Garg, P. and Fetzer, T. (2025). Causal claims in economics. *arXiv preprint arXiv:2501.06873*.
- Iacus, S. M., King, G., and Porro, G. (2012). Causal inference without balance checking: Coarsened exact matching. *Political Analysis*, 20(1):1–24.
- LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *American Economic Review*, 76(4):604–620.